

Mathematical Methods — Units 3 & 4

Differentiation · Practice Examination 1 — Worked Solutions

Full worked solutions with method marks indicated.

1. Differentiate $y = x^3 - 4x^2 + 7x - 2$. (3 marks)
Solution. $dy/dx = 3x^2 - 8x + 7$.
2. Find dy/dx for $y = (2x + 1)^5$. (3 marks)
Solution. Chain rule: $dy/dx = 5(2x + 1)^4 \times 2 = 10(2x + 1)^4$.
3. Find the equation of the tangent to $y = x^2 - 3x$ at the point where $x = 2$. (4 marks)
Solution. At $x = 2$, $y = 4 - 6 = -2$, so the point is $(2, -2)$.
 $dy/dx = 2x - 3$, which equals 1 at $x = 2$.
Tangent: $y - (-2) = 1(x - 2)$, i.e. $y = x - 4$.
4. Differentiate $y = x e^x$ and find the x -coordinate of the stationary point. (3 marks)
Solution. Product rule: $dy/dx = e^x + x e^x = e^x(1 + x)$.
Stationary when $1 + x = 0$, so $x = -1$.
5. Differentiate $y = \ln(3x^2 + 1)$. (3 marks)
Solution. Chain rule: $dy/dx = 6x / (3x^2 + 1)$.
6. For the curve $y = x^3 - 6x^2 + 9x$, find the coordinates and nature of the stationary points. (4 marks)
Solution. $dy/dx = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$.
Stationary points at $x = 1$ and $x = 3$.
 $(1, 4)$ is a local maximum; $(3, 0)$ is a local minimum.